AI Derma

Project Setup & Run Guide

Service Port Summary

Service	Port	URL
Frontend (Vite / React)	5173	http://localhost:5173
Backend API (.NET)	5127	http://localhost:5127
AI Model API (FastAPI)	8001	http://localhost:8001
Expert System API (FastAPI)	8002	http://localhost:8002
SQL Server	1433	(default)

1. Backend API — .NET 10 / C#

Technologies Required

Technology	Details
.NET 10 SDK	Primary runtime and build tool
SQL Server Express	Relational database (MSSQL)
Entity Framework Core 10.0.5	ORM for database access
JWT Authentication	Secure token-based auth
Cloudinary	Cloud image storage
Swashbuckle / Swagger	API documentation (v10.1.7)
CloudinaryDotNet 1.29.1	Cloudinary .NET SDK
dotenv.net 4.0.2	Environment variable loader

How to Run

Step 1 – Install .NET 10 SDK

```
winget install Microsoft.DotNet.SDK.10
```

Step 2 – Install SQL Server Express

```
winget install Microsoft.SQLServer.Express
```

Step 3 – Update Connection String

Open **backend-api/AI-Derma/AI-Derma/appsettings.json** and replace **DESKTOP-EMA2VNQ** with your machine name in the DefaultConnection string.

Step 4 – Run Database Migrations

```
cd backend-api/AI-Derma/AI-Derma
dotnet restore
dotnet ef database update --project ../AI-Derma.Infrastructure
```

Step 5 – Build & Run

```
cd backend-api/AI-Derma/AI-Derma
dotnet restore
dotnet build
dotnet run
# Runs on http://localhost:5127
# Swagger UI: http://localhost:5127/swagger
```

Required Configuration (appsettings.json)

Technology	Details
ExpertSystem:BaseUrl	http://127.0.0.1:8002
PredictionService:BaseUrl	http://127.0.0.1:8001
JWT:SecretKey	Your JWT signing secret
Groq:ApiKey	Your Groq API key
CloudinarySettings	CloudName, ApiKey, ApiSecret

2. AI Model API — Python + FastAPI + TensorFlow

Technologies Required

Technology	Details
Python 3.8+	Runtime language
FastAPI	Web framework for the API
TensorFlow / Keras	Deep learning inference
EfficientNetB0	Pre-trained neural network backbone
Uvicorn	ASGI server
Pillow	Image processing
NumPy	Numerical computing
OpenCV (cv2)	Computer vision utilities

How to Run

Step 1 – Create & activate virtual environment

```
cd ai-model/kaggle\ run
python -m venv venv
venv\Scripts\activate          # Windows
source venv/bin/activate        # macOS/Linux
```

Step 2 – Install dependencies

```
pip install fastapi uvicorn tensorflow keras pillow numpy opencv-python
```

Step 3 – Start the server

```
uvicorn app:app --host 0.0.0.0 --port 8001 --reload
# API:  http://localhost:8001
# Docs: http://localhost:8001/docs
```

Model Details

Model file: **derma_model.keras** — EfficientNetB0-based, input size 224x224 RGB. Classifies 10 skin conditions: Acne, Basal Cell Carcinoma, Benign Keratosis-like Lesions, Eczema, Melanocytic Nevi, Melanoma, Psoriasis, Seborrheic Keratoses, Tinea Ringworm Candidiasis, Warts Molluscum.

3. Expert System API — Python + FastAPI

Technologies Required

Technology	Details
Python 3.8+	Runtime language
FastAPI	Web framework for the API
Uvicorn	ASGI server
Pydantic	Data validation
expert_logic.py	Custom rule-based logic module
expertsystem.py	Core expert system implementation

How to Run

Step 1 – Create & activate virtual environment

```
cd testexpsystem
python -m venv venv
venv\Scripts\activate          # Windows
source venv/bin/activate        # macOS/Linux
```

Step 2 – Install dependencies

```
pip install fastapi uvicorn pydantic
```

Step 3 – Start the server

```
uvicorn main:app --host 0.0.0.0 --port 8002 --reload
# API:  http://localhost:8002
# Docs: http://localhost:8002/docs
```

4. Frontend Web — React + Vite

Technologies Required

Technology	Details
Node.js 18+	JavaScript runtime (with npm)
React 19.2.4	UI library
Vite 8.0.4	Build tool with hot-module reload
React Router DOM 7.14.1	Client-side navigation
Axios 1.15.0	HTTP client
ESLint 9.39.4	Code linting
@vitejs/plugin-react 6.0.1	Vite plugin for React

How to Run

Step 1 – Install Node.js from https://nodejs.org (LTS)

Step 2 – Create environment file

Create **.env** inside **frontend-web/react-project-app/** with:

```
VITE_API_BASE_URL=http://localhost:5127
```

Step 3 – Install dependencies & run

```
cd frontend-web/react-project-app
npm install
npm run dev
# Runs on http://localhost:5173
```

Other useful scripts

Technology	Details
npm run build	Build for production
npm run preview	Preview the production build
npm run lint	Run ESLint checks

5. Mobile App — Flutter

Technologies Required

Technology	Details
Flutter SDK 3.0.0+	Cross-platform mobile framework
Dart SDK 3.0.0+	Language runtime
Android SDK	Required for Android targets
Xcode (macOS only)	Required for iOS targets
provider ^6.1.1	State management
go_router ^13.0.0	Navigation
google_fonts ^6.1.0	Custom font support
image_picker ^1.0.7	Camera / gallery image input
http ^1.2.0 / dio ^5.4.3	HTTP networking
shared_preferences ^2.2.3	Local key-value storage
cached_network_image ^3.3.1	Network image caching

How to Run

Step 1 – Install Flutter (add to PATH, then verify)

```
flutter doctor
```

Step 2 – (Android) Accept licenses

```
flutter doctor --android-licenses
```

Step 3 – (iOS / macOS only) Install CocoaPods

```
sudo gem install cocoapods
cd ios && pod install && cd ..
```

Step 4 – Get dependencies & run

```
cd mobile-flutterr/ai_derma_new
flutter pub get
flutter run
```

Build commands

Technology	Details
flutter build apk --release	Android release APK
flutter build ios --release	iOS release (macOS only)
flutter devices	List connected devices
flutter run -d <device_id>	Run on a specific device

6. Recommended Startup Order

Technology	Details
Terminal 1	Start SQL Server (Windows Services or SSMS)
Terminal 2	AI Model API — uvicorn on port 8001
Terminal 3	Expert System API — uvicorn on port 8002
Terminal 4	Backend API — dotnet run on port 5127
Terminal 5	Frontend Web — npm run dev on port 5173
Separate	Mobile App — flutter run

Troubleshooting

Backend API

```
dotnet ef database drop
dotnet ef database update
# Also check appsettings.json server name
```

Python APIs

```
python --version          # must be 3.8+
pip install -r requirements.txt
```

Frontend

```
rm -rf node_modules
npm install
rm -rf .vite
npm run dev
```

Mobile

```
flutter clean
flutter pub get
flutter run -d
```